# Heterodyne Density Lab

Scientific Model, Numerical Methods, and Software Reference

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**Abstract**

Heterodyne Density Lab models a one-dimensional photodiode-array interferometer for a cylindrically symmetric plasma. A finite-support density profile is Abel integrated, converted to optical phase, coherently averaged over finite detector areas, and measured using an ideal calibrated heterodyne I/Q chain. Detector and shot noise are added to the quadratures before phase recovery. A rank-protected Pretzler basis with tempered detector weighting reconstructs the radial density, followed by an even penalized cubic B-spline. Separate scripts perform baseline simulation, spatial-resolution, power, noise-decomposition, and basis-order studies. One global estimator is trained per detector layout and held fixed through physical sweeps. Portable CFFI kernels accelerate quadrature and phase operations; Python manages configuration, optimized linear algebra, statistics, and output. This manual describes the implemented mathematics, deliberate consolidations from the original MATLAB package, validation, and limitations.

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# 1 Purpose, scope, and relationship to the original package

The package is a synthetic diagnostic-design tool, not a general tomography solver or a wave-propagation simulation. The physical measurement model follows the supplied MATLAB package and its scientific documentation: the observable is phase recovered from a finite-area heterodyne phasor, rather than an arbitrary noisy chord integral. The original MATLAB repository remains separate.

The open implementation preserves the transformed trapezoidal Abel rule, finite-area quadrature, detector tables and ideal I/Q gains, constant/cosine Pretzler basis, tempered weighting, global estimator training, and all four study types. It uses one shared forward engine and one shared reconstruction engine rather than one compatibility wrapper for every old file.

The source MATLAB package offered proprietary-toolbox and open penalized-spline paths. This release makes the latter canonical. It does not claim numerical identity to MATLAB's toolbox smoothing-spline auto-selection. NumPy PCG64 replaces MATLAB's random generator; equal integer seeds do not give equal cross-language normal variates. These are explicit implementation choices, not hidden physical-model changes.

# 2 Installation, execution, and reproducibility

## 2.1 Environment and build

The repository README and three installation guides provide fresh-system instructions for standard 64-bit CPython 3.12/3.13, Poetry 2.x, and a native C compiler. Use Clang on macOS, GCC on Linux, and MSVC on Windows. Python, the compiler target, and the generated extension must have matching architectures. A Poetry virtual environment is a Python environment, not an operating-system VM.

Poetry installs the locked dependencies and root namespace. The documented explicit editable-install command then invokes the setuptools backend and its CFFI hook inside that environment. The build declares CFFI separately from the runtime dependency because isolated source builds need it too. No MATLAB, Anaconda, GPU toolkit, mandatory OpenMP, or C++ application code is required.

## 2.2 Five individual scientific scripts

Edit `config/default_design.py` and `config/study_config.py`, then run

```
poetry run python scripts/main_density_design.py
poetry run python scripts/study_density_scale_length.py
poetry run python scripts/study_detector_power_tradeoff.py
poetry run python scripts/study_detector_noise_contribution.py
poetry run python scripts/study_pretzler_basis_order.py
```

Use the repository root as the working directory. A shared loader resolves the trusted Python configuration files relative to each script's repository, with no per-script path mutation. An explicit `--quick` option reduces the study preset for installation checks and records that choice; it is not the default scientific study. `--refresh-tuning` recomputes matching cached training.

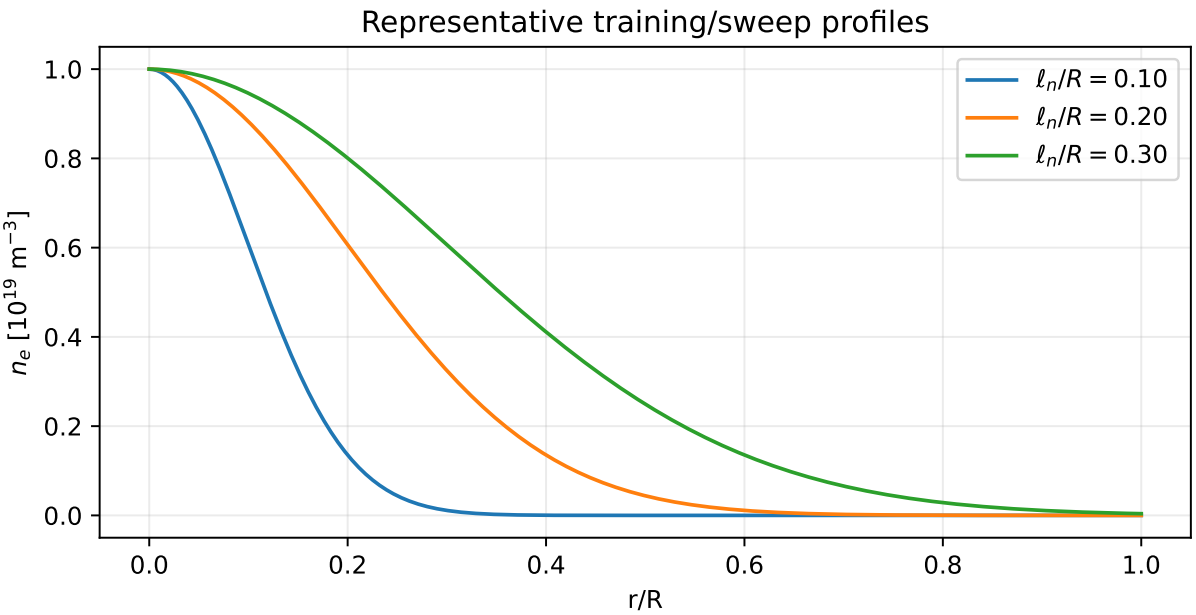



Figure 1: Representative prescribed profiles used in training and sweeps, with $n_a = 2$, $n_b = 0$, and common peak density.

Runs create unique UTC-stamped directories under `output/`. They save configuration/environment JSON, CSV results with units, figures, and timing. Single designs also save numeric NPZ ensembles. JSON uses `null` for nonfinite diagnostics; CSV invalid-power entries are `nan`. By default, figures are saved without a display. Set `cfg.plots.show=True` for windows. Documentation figures are deliberate exports and are not overwritten by runs.

## 3 Coordinate convention and plasma density

The laser propagates along $z$. The detector row spans $x$ and the long direction of an element is $y$. The plasma is symmetric in the $x$–$z$ plane and uniform in $y$:

$$r = \sqrt{x^2 + z^2}. \tag{1}$$

All lengths are meters, densities m$^{-3}$, powers watts, and frequencies hertz. For $0 \le r \le R$, the prescribed density is

$$n(r) = (n_0 - n_b) \exp\left[-\frac{(r/\ell_n)^{n_a}}{n_a}\right] + n_b; \qquad n(r) = 0 \quad (r > R). \tag{2}$$

Here $n_b$ is a profile parameter; it does not impose a reconstruction boundary condition or force $n(R) = n_b$. The default values are $R = 0.10$ m, $n_0 = 10^{19}$ m$^{-3}$, $n_b = 0$, $\ell_n = 0.025$ m, and $n_a = 2$. The study parameter $\ell_n$ is distinct from the local scale length $L_n(r)$.

Table 1: Built-in one-row detector geometry, following the source model and manufacturer specifications.

| Model | Elements | Active size [mm] | Pitch [mm] |
|---|---|---|---|
| G12430-016D | 16 | $0.45 \times 1.00$ | 0.50 |
| G12430-032D | 32 | $0.20 \times 1.00$ | 0.25 |
| G12430-046D | 46 | $0.20 \times 1.00$ | 0.25 |

# 4 Abel forward transform and quadrature

For impact parameter $x$,

$$h(x) = 2 \int_{|x|}^{R} \frac{n(r) r}{\sqrt{r^2 - x^2}} \, \mathrm{d}r. \tag{3}$$

With $r = \sqrt{x^2 + s^2}$, the endpoint singularity disappears:

$$h(x) = 2 \int_{0}^{\sqrt{R^2 - x^2}} n(\sqrt{x^2 + s^2}) \, \mathrm{d}s, \qquad h(x) = 0 \quad (|x| \geq R). \tag{4}$$

For $N_q$ nodes, let $t_k = k/(N_q - 1)$ and $s_k = s_{\max} t_k$. The source's trapezoidal rule is retained,

$$h(x) \simeq \frac{2 s_{\max}}{N_q - 1} \sum_{k=0}^{N_q - 1} w_k n(\sqrt{x^2 + s_k^2}), \qquad w_0 = w_{N_q - 1} = \tfrac{1}{2}, \tag{5}$$

with other $w_k = 1$. The default is 1000 nodes. The C kernel streams through nodes without allocating a radius-by-node matrix. For $n_a = 2$, the exponent can be computed from $x^2 + s^2$ directly, avoiding a square root in the density exponent.

The finite-radius Gaussian has an analytic reference for $n_b = 0$, $n_a = 2$:

$$h(x) = n_0 e^{-x^2/(2\ell_n^2)} \sqrt{2\pi} \ell_n \operatorname{erf}\left( \frac{\sqrt{R^2 - x^2}}{\sqrt{2} \ell_n} \right). \tag{6}$$

This and the constant-profile expression $h = 2 n_0 \sqrt{R^2 - x^2}$ test the kernel independently of the inverse problem.

# 5 Imaging, beams, aperture, and finite detectors

## 5.1 Signed magnification and detector geometry

A signed magnification maps $x_I = M x_O$, $y_I = M y_O$. Detector quadrature is in the image plane; beam/aperture functions use $x_O = x_I / M$, $y_O = y_I / M$. $M$ must be finite and nonzero. Both signs are supported.

Custom geometry requires element count, active width/height, pitch, and image center. Pitch cannot be smaller than active width. User selections and table IDs are physical indices $1, \ldots, N_e$; internal NumPy indices are zero-based.

### 5.2 Probe/reference shape and port normalization

The unblocked Gaussian intensity shape is

$$G_P = \exp\left[-2\frac{(x_O - x_P)^2 + (y_O - y_P)^2}{w_P^2}\right]. \tag{7}$$

$w_P$ is the field $1/e$ radius, equivalently the intensity $1/e^2$ radius. A circular port has mask $A = 1$ inside its radius and zero outside. The transmitted probe shape is $S_P = G_P A$. A matched reference uses $S_R = S_P$; `unclipped_gaussian` uses an independent Gaussian without that port mask. Reference waist and center can differ in the latter mode.

Total shape integrals in the image plane are $T_P = |M|^2 \int S_P \, \mathrm{d}A_O$ and $T_R = |M|^2 \int S_R \, \mathrm{d}A_O$. A centered Gaussian port integral is evaluated analytically; for an offset port the source's polar quadrature is retained, with at least 80 radial Gauss nodes and 256 uniform angular nodes. An unclipped Gaussian has total $\pi w_R^2/2$ in object coordinates.

### 5.3 Finite-area quadrature and exact transverse reduction

Tensor Gauss–Legendre quadrature uses the default 20 nodes per direction on each physical rectangular element. Let $a_{jkl}$ be the image-plane area weight and $S_{P,jkl}$, $S_{R,jkl}$ the beam shapes. The cross weight is

$$c_{jkl} = a_{jkl}\sqrt{S_{P,jkl} S_{R,jkl}}. \tag{8}$$

The element fractions are

$$\eta_{P,j} = \frac{\sum a_{jkl} S_{P,jkl}}{T_P}, \quad \eta_{R,j} = \frac{\sum a_{jkl} S_{R,jkl}}{T_R}, \quad \eta_{X,j} = \frac{\sum c_{jkl}}{\sqrt{T_P T_R}}. \tag{9}$$

The effective signed coordinate is the cross-weight centroid. Elements are retained when selected, passing the probe transmission threshold (default 0.05), and above the source's relative numerical cross-weight floor.

Because phase depends only on $x_O$, define $v_{jk} = \sum_l c_{jkl}$. Then $\sum_{kl} c_{jkl} e^{i\phi(x_{jk})} = \sum_k v_{jk} e^{i\phi(x_{jk})}$ exactly on the same quadrature rule. This removes repeated $y$ work without changing aperture clipping, reference mismatch, or the finite-area observable. Absolute transverse nodes are then deduplicated across elements. The MATLAB source already deduplicated within each element; the improvement extends that reuse.

## 6 Finite-area heterodyne phase measurement

The optical phase associated with the chord integral is

$$\phi(x) = -r_e \lambda h(x), \qquad r_e = 2.8179403262 \times 10^{-15} \text{ m}. \tag{10}$$

The plasma and vacuum cross phasors are

$$Z_{p,j} = \int_{D_j} \sqrt{S_P S_R}\, e^{i\phi} \, \mathrm{d}A_I, \qquad Z_{0,j} = \int_{D_j} \sqrt{S_P S_R} \, \mathrm{d}A_I. \tag{11}$$

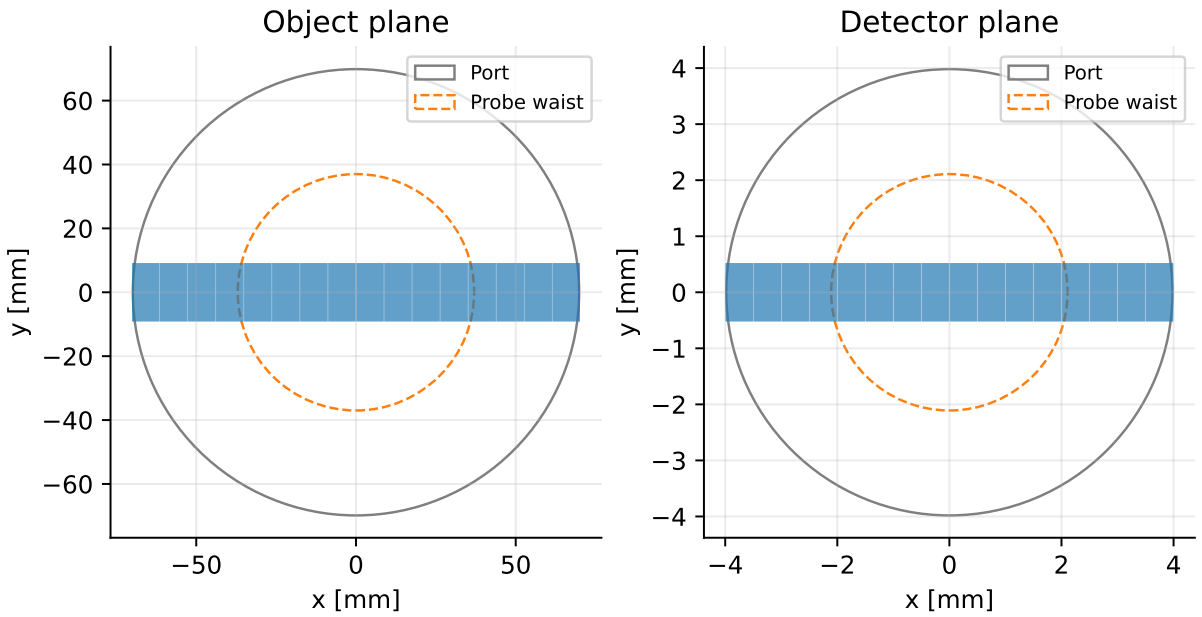



Figure 2: Default object/image geometry. Solid detector rectangles show selected physical elements, the gray circle the port, and the dashed circle the probe waist. Ideal imaging is assumed.

The ideal calibration gives

$$\Delta\phi_j = \arg(Z_{p,j} Z_{0,j}^*), \qquad h_j^{(\phi)} = -\frac{\Delta\phi_j}{r_e \lambda}. \tag{12}$$

The vacuum phasor is real and nonnegative because this model has no static wavefront phase. Phase recovery returns the principal value without unwrapping. Zero-overlap channels have no defined measurement and are excluded.

For weak phase,

$$h_j^{(\mathrm{lin})} = \frac{\int_{D_j} \sqrt{S_P S_R}\, h \, \mathrm{d}A_I}{Z_{0,j}} \tag{13}$$

approximates Eq. (12). Both are returned with their difference. The coherence $\gamma_j = |Z_{p,j}|/Z_{0,j}$ diagnoses within-element cancellation. A large difference between exact phase and the linearized average indicates model mismatch for the linear inverse problem, even when the synthetic forward phasor remains well-defined.

# 7 Absolute optical power, detector data, and noise

## 7.1 Power split and carrier

Let $P$ be the combined full detector-plane beam power, not merely the power intercepted by the array. With probe fraction $f$ (default 1/2), $P_{P,\mathrm{tot}} = fP$ and $P_{R,\mathrm{tot}} = (1-f)P$. Element powers are $P_{P,j} = fP\eta_{P,j}$ and $P_{R,j} = (1-f)P\eta_{R,j}$. Responsivity $\mathcal{R}$ gives $I_{\mathrm{DC},j} = \mathcal{R}(P_{P,j} + P_{R,j})$. For $\zeta_j = Z_{p,j}/\sqrt{T_P T_R}$, the complex IF current is

$$C_j^{\mathrm{IF}} = 2\mathcal{R}P\sqrt{f(1-f)}\,\zeta_j. \tag{14}$$

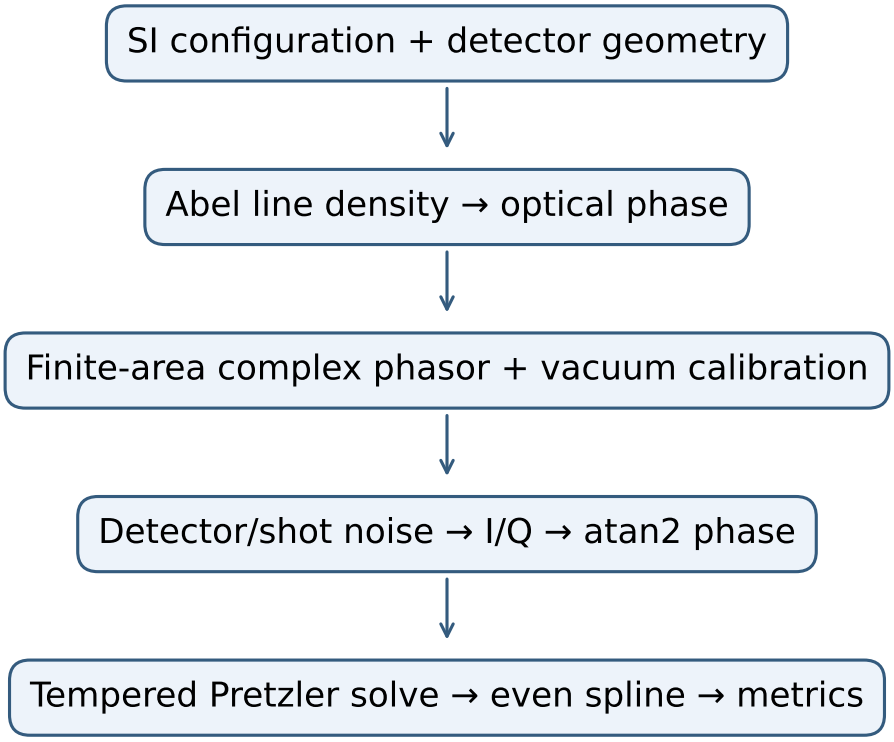



Figure 3: Implemented calculation sequence. Every study calls the same forward and reconstruction engines.

The inherited ideal demodulator convention is

$$C_j = \frac{\sqrt{2}}{\pi} C_j^{\mathrm{IF}}. \tag{15}$$

It changes the I/Q amplitude, not the carrier phase. The physical motivation for ideal heterodyne detection and quadrature extraction is discussed by Davis [1]; this implementation intentionally follows the normalization specified in the supplied package.

### 7.2 Detector specifications

Typical 1.55 $\mu$m responsivity is 0.95 A/W (conservative: 0.85). The 16-element model uses typical/conservative NEP $2/4 \times 10^{-14}$ W/$\sqrt{\mathrm{Hz}}$ and cutoff 30/15 MHz. The 32/46-element models use NEP $1/3 \times 10^{-14}$ and cutoff 60/25 MHz. Dark current, capacitance, and shunt resistance are retained as specifications, but are not independently added as noise contributions on top of the detector-equivalent NEP. These values follow the supplied model and the manufacturer's data [2].

Custom detectors require positive responsivity, NEP, and cutoff, with optional additional fields and a linearity limit. Wavelength-dependent response is not interpolated from a graph; built-in responsivity/NEP remain peak-wavelength approximations. A request outside the 0.9–1.7 $\mu$m range triggers a warning. The cutoff is diagnostic only, with the stated datasheet test conditions, not a complete TIA/RF transfer function.

### 7.3 Noise ASD and quadrature samples

The input current-noise ASD is

$$i_{n,j} = \sqrt{(\mathcal{R}\,\mathrm{NEP})^2 + 2eI_{\mathrm{DC},j}}, \tag{16}$$

with the shot term omitted when configured. Each real baseband quadrature has ASD $(2/\pi)i_{n,j}$ and standard deviation

$$\sigma_{i,j} = \frac{2}{\pi} i_{n,j}\sqrt{B}, \tag{17}$$

where $B$ is post-demodulation ENBW (default 1 MHz). Independent normal samples give

$$I_j = \Re C_j + \sigma_{i,j}\xi_{I,j}, \quad Q_j = \Im C_j + \sigma_{i,j}\xi_{Q,j}, \quad h_j = -\frac{\mathrm{atan2}(Q_j, I_j)}{r_e\lambda}. \tag{18}$$

The high-SNR diagnostics are

$$\sigma_{\phi,j} \simeq \frac{\sigma_{i,j}}{|C_j|}, \qquad \sigma_{h,j} = \frac{\sigma_{\phi,j}}{r_e\lambda}, \qquad \mathrm{SNR}_{C,j} = |C_j|/\sigma_{i,j}. \tag{19}$$

They supply detector weights but do not replace the noisy I/Q generator. The preferred carrier SNR is 5; low-SNR and IF-plus-ENBW/cutoff warnings preserve the limits of this analytic approximation.

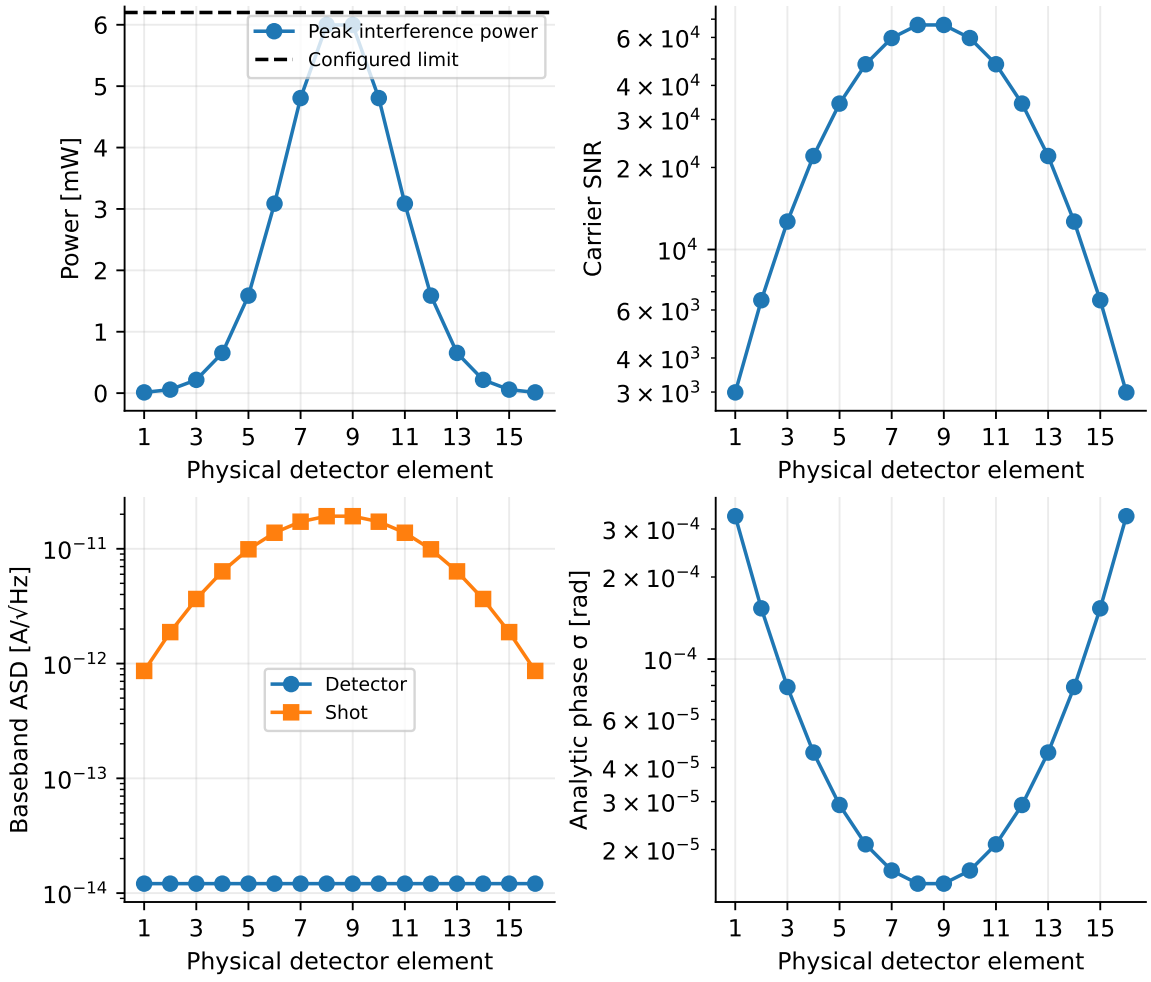



Figure 4: Baseline peak power, carrier SNR, detector/shot baseband ASD, and analytic phase uncertainty. These are engineering-model diagnostics, not receiver calibration measurements.

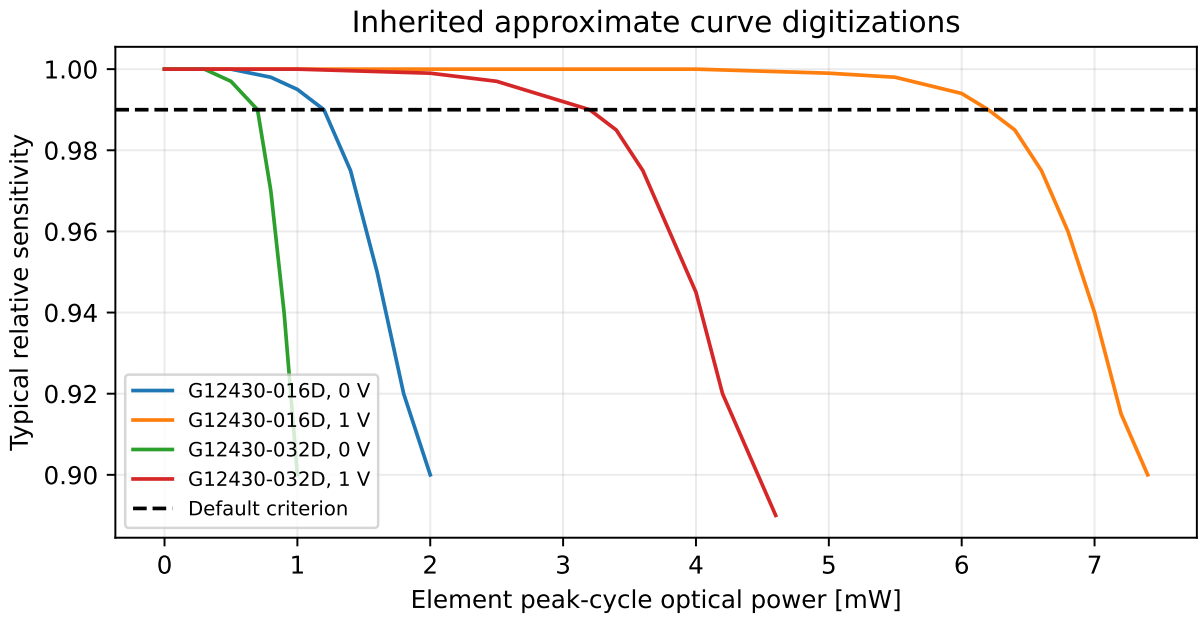



Figure 5: Approximate typical sensitivity curves used by the model. The 32/46-element models share the same digitization. The horizontal dashed line is the default criterion.

## 8 Typical-linearity constraint and admissible power

Peak instantaneous heterodyne-cycle element power is conservatively evaluated using the vacuum overlap:

$$P_{\mathrm{peak},j} = P\left[f\eta_{P,j} + (1-f)\eta_{R,j} + 2\sqrt{f(1-f)}\eta_{X,j}\right] \equiv Pa_j. \tag{20}$$

The source's approximate digitizations of typical relative-sensitivity curves at 0/1 V are preserved. The default criterion is 99% sensitivity at 1 V. A safety factor in $(0, 1]$ scales the per-element allowable power $P_L$. No interpolation beyond the digitized graph is used. Plateau values are collapsed using their highest-power point when inverting sensitivity to power. The combined beam-power ceiling is

$$P_{\max} = \min_{j:a_j>0} P_L/a_j. \tag{21}$$

The constraint covers every illuminated physical element, including unselected channels. It is enforced even in noiseless diagnostic mode. The default geometry gives about 0.05169 W allowable total power; the 0.05 W source baseline is close to this configured boundary. The power study retains rows beyond the boundary but marks them invalid and leaves reconstruction errors as NaN.

The curves are typical operating estimates, not guaranteed damage thresholds. The manufacturer's spot/bias conditions differ from an arbitrary Gaussian footprint and receiver. Receiver amplifier, mixer, TIA, and ADC saturation are not included. Figure 5 plots the inherited numeric digitizations; it does not reproduce manufacturer artwork.

# 9 Pretzler inversion and tempered detector weighting

## 9.1 Basis and finite-area matrix

The derivative-free basis follows the supplied Pretzler implementation [3]:

$$F_0(r) = 1, \qquad F_n(r) = 1 - (-1)^n \cos(n\pi r/R), \quad n \geq 1. \tag{22}$$

The estimated profile is $n(r) = \sum_b A_b F_b(r)$ using the configured lower/upper orders and optional constant. Each basis is Abel integrated with the same transformed trapezoidal rule. The inverse matrix is

$$K_{jb} = \sum_k \frac{v_{jk}}{\sum_k v_{jk}} \mathcal{A}[F_b](x_{jk}), \tag{23}$$

for retained channels. Its rows represent the *linearized* cross-weighted finite-area observable, even though the simulation supplies exact phase-derived measurements. No nonlinear inverse problem is introduced in this release.

A maximum-order basis bank is cached per radius and quadrature resolution. Constant-basis line integrals are exact chord lengths. Higher cosines use a recurrence with one initial cosine per integration node. Lower-order candidates select leading columns instead of regenerating optics or basis integrals.

## 9.2 Rank protection and stable solve

If coefficients outnumber retained measurements, or a leading-column matrix is rank deficient, the largest full-rank leading prefix is selected when rank limiting is enabled. Otherwise a clear error is raised. No extra basis functions are added automatically because more measurements exist. Symmetric arrays may have many physical channels but fewer independent radial observations. Requested/used order, rank reduction, rank, and condition numbers are returned.

For detector-derived uncertainties, normalized row factors are

$$d_j = (\sigma_{h,j} / \operatorname{median} \sigma_h)^{-\alpha}, \qquad 0 \leq \alpha \leq 1. \tag{24}$$

The solve minimizes

$$\|D(KA - h)\|_2^2, \qquad D = \operatorname{diag}(d_j). \tag{25}$$

Thus $\alpha = 0$ is OLS and $\alpha = 1$ full inverse-variance WLS. A small SVD-based least-squares operator is computed once per candidate and applied to all realizations. This changes neither the least-squares objective nor the trained weight exponent. A matrix can be full rank yet poorly conditioned; warnings begin at the configurable default $10^6$.

# 10 Symmetric penalized cubic B-spline

The continuous reconstruction fits in

$$u = (|x|/R)^2, \tag{26}$$

so the resulting radial function is even and has zero central derivative. Pretzler values at effective detector positions supply fit points. Additional support at 0 and $\pm R$ comes from the *same Pretzler*

*expansion*, not the true profile or a known background value. Opposite-side/duplicate $u$ values are merged after rounding to 12 decimals; multiplicities $c_i$ weight the fit. The source requires at least four unique effective detector radii before adding center/edge support; that guard is preserved.

Let $B$ be an open-uniform cubic basis at the unique $u_i$, $B_w = C^{1/2}B$, $y_w = C^{1/2}y$, and $D_2$ the second difference of the coefficient identity matrix. For penalty $\lambda_s$, the source fallback solves

$$\left(B_w^T B_w + \lambda_s D_2^T D_2 + 10^{-12} I\right) b = B_w^T y_w. \tag{27}$$

The basis count is at most the configured default 10 and the number of unique support radii, with a minimum of four. The fixed ridge is retained exactly. Scaling $y_w$ by its maximum absolute value protects SI-scale densities and is reversed after the solve. Optional nonnegative clipping is off by default.

If no fixed penalty is supplied, 80 logarithmic candidates from $10^{-8}$ to $10^4$ are compared by per-realization generalized cross-validation:

$$\mathrm{GCV}(\lambda_s) = \frac{\|B_w b - y_w\|^2 / N_u}{\max(1 - \operatorname{tr} H/N_u, \sqrt{\epsilon})^2}, \quad H = B_w (B_w^T B_w + \lambda_s D_2^T D_2 + 10^{-12} I)^{-1} B_w^T. \tag{28}$$

Basis, merge, solve, and hat operators are cached and applied in batches; each realization still selects its own minimizing penalty. This is the same algebraic objective as the MATLAB open fallback, tested against an independent sample-wise implementation. Floating-point reductions need not match bitwise.

## 11 Error metrics and statistical summaries

The full-radius density metric samples a uniform $0 \le r \le R$ grid:

$$E_n = \frac{\sqrt{\frac{1}{N_r}\sum_i [n_{\mathrm{rec}}(r_i) - n_{\mathrm{true}}(r_i)]^2}}{\max_i n_{\mathrm{true}}(r_i)}. \tag{29}$$

Point-only and sampled-radius errors are also reported for Pretzler/spline profiles; the primary study metric is full-radius spline error.

The local scale length is $L_n = |n/(\,\mathrm{d}n/\,\mathrm{d}r)|$. Derivatives use centered finite differences in the interior and one-sided endpoint differences on the uniform grid, matching the source's gradient convention. Gradients below $10^{-12}$ of each profile's maximum gradient are treated as zero, giving infinite $L_n$. The bounded error is

$$e_L = |L_{n,\mathrm{rec}} - L_{n,\mathrm{true}}| / (L_{n,\mathrm{rec}} + L_{n,\mathrm{true}}). \tag{30}$$

Both-infinite and both-zero lengths receive zero error; one-sided infinity receives one. The reported $E_L$ is RMS over $0 < r \le R$ and lies in $[0, 1]$. This convention avoids divergent relative errors near stationary gradients.

Summary statistics include mean, sample standard deviation (zero for one realization), median, and linearly interpolated 5th/95th percentiles. A single realization is not an uncertainty estimate, even when it is a source default. Study CSVs retain standard deviations even where presentation figures omit bars.

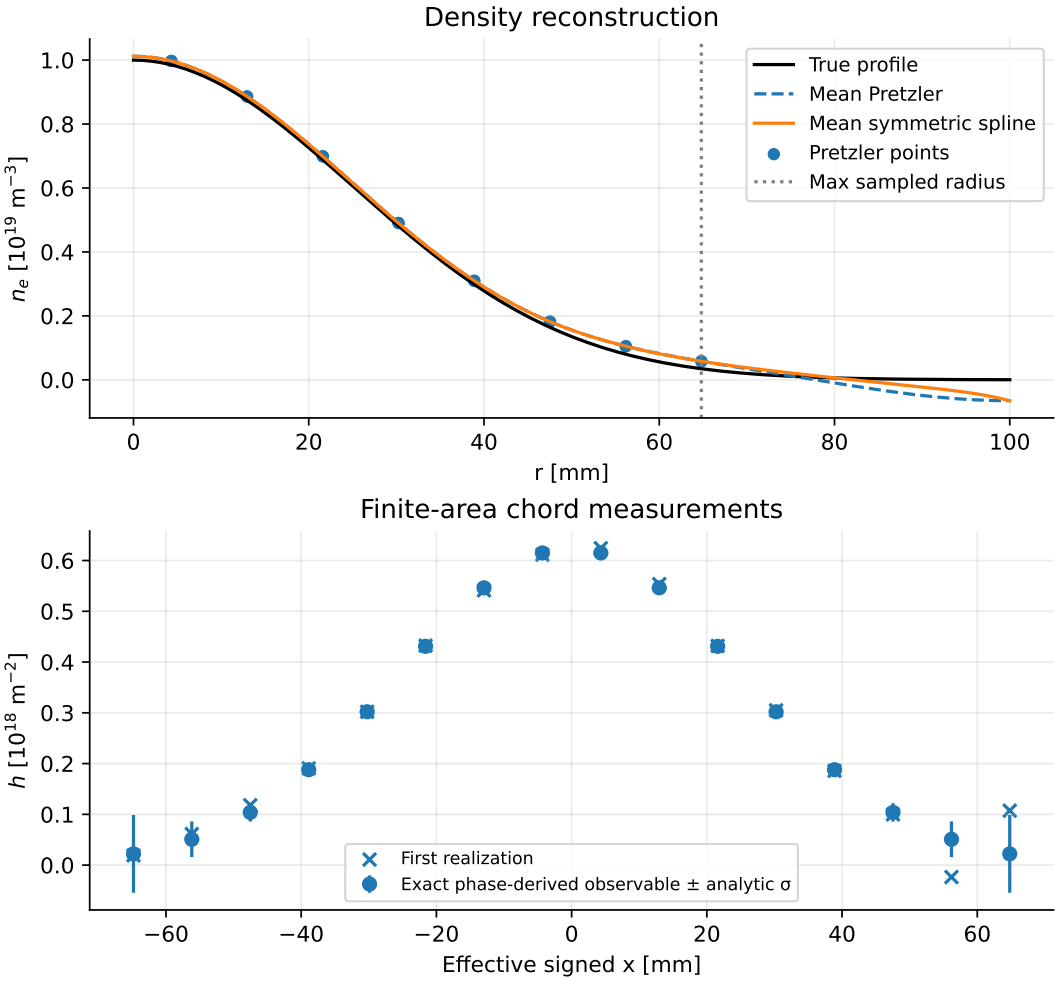



Figure 6: Default single-design reconstruction and phase-derived chord data. This run contains one noise realization. The vertical line marks the largest effective detector radius; outer-radius behavior is extrapolation through the chosen basis and spline support.

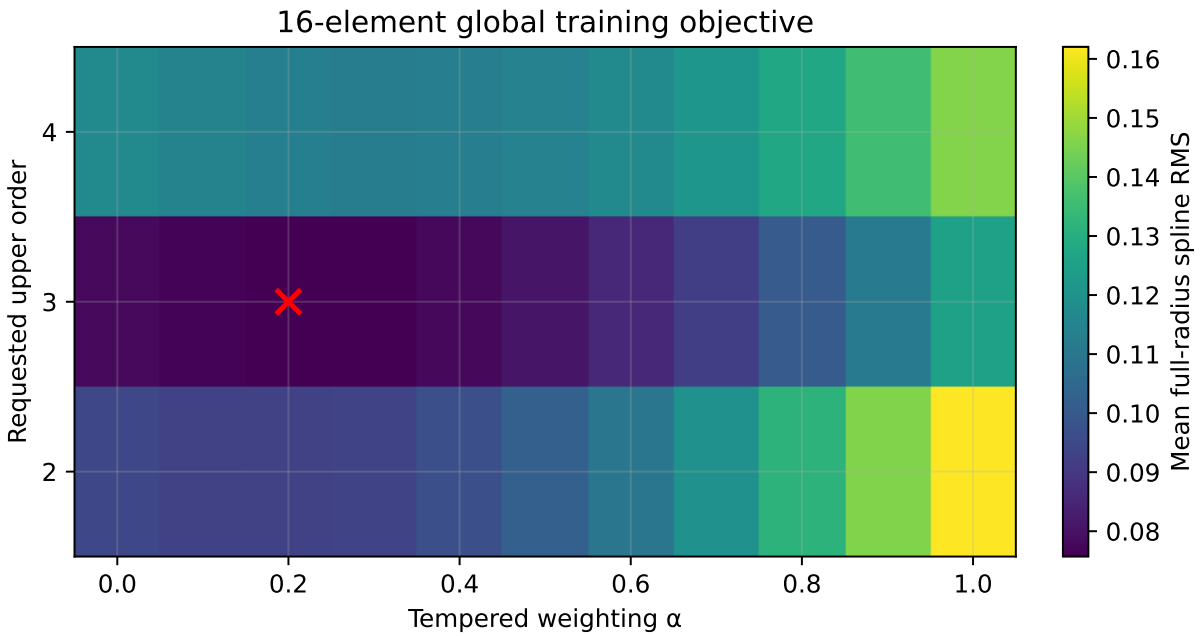



Figure 7: Training score surface for the baseline 16-element layout. The selected pair is requested upper order 3 and $\alpha = 0.2$ for this run. Random-stream and spline choices are part of the estimator definition.

## 12 Noise modes and globally fixed estimator selection

Four modes are supported. `detector_limited` adds independent noise to I/Q before phase recovery and uses detector-derived weights. `detector_weighted_noiseless` keeps exact phase with those weights. `noiseless` uses exact phase with OLS. The legacy `relative` mode adds multiplicative Gaussian noise directly to line density and uses OLS; it is retained as a diagnostic and is not the default physical heterodyne noise model.

Each detector layout is tuned once at the nominal 0.05 W. Training profiles are

$$\ell_n/R \in \{0.10, 0.15, 0.20, 0.25, 0.30\}. \tag{31}$$

Candidates use upper orders $(2, 3, 4)$ and $\alpha = 0, 0.1, \ldots, 1$. Default training count is 100 realizations per profile. All candidate pairs use one common I/Q ensemble per profile. The aggregate score is the mean configured metric across profiles and realizations; default is full-radius spline RMS. The selected pair is fixed before physical sweeps.

Tie traversal follows the source's column-major order: lower alpha first, then lower order. Rank-protected candidate orders can share the same used prefix. An optimum on the upper search boundary triggers a warning. Expanding the search range is a conscious scientific choice, not a hidden script override.

# 13 Central detector layouts and design studies

## 13.1 Source layout coverage

All nine supplied layouts use G12430-016D, $M = 0.057$, and zero image center:

| Layout | Physical element IDs |
|---|---|
| 16 elements | 1–16 |
| 8, full coverage | 1,3,5,7,9,11,13,15 |
| 8, gradient paired | 5,6,7,8,9,10,11,12 |
| 6, broad coverage | 3,5,7,9,11,13 |
| 6, inner gradient paired | 5,6,7,8,9,10 |
| 5, inner-mid coverage | 5,7,9,11,13 |
| 5, gradient plus outer | 3,5,7,11,13 |
| 4, gradient band | 5,7,11,13 |
| 4, inner gradient | 5,7,9,11 |

Other built-in/custom detectors can be substituted explicitly in study configuration. Changing selected channels never removes physical-element power constraints.

## 13.2 Spatial-resolution study

Nine $\ell_n/R$ values span 0.10–0.30, with 1000 realizations per layout/profile. Density and bounded local-scale-length errors are recorded. Figures in the run output group layouts by digitized-channel count. The manual shows three representative layouts for readable comparisons; the CSV contains all nine.

## 13.3 Optical-power tradeoff

The source grid spans $10^{-3}$–$10^{-1}$ W at eight points per decade and profiles $\ell_n/R = 0.10, 0.20, 0.30$. The fixed globally tuned estimator is retained. Default one-realization curves are accompanied by validity, carrier SNR, and linearity utilization. Forward phasors do not depend on total power and are reused across this sweep. Inadmissible powers are not reconstructed while linearity enforcement is enabled. Disabling enforcement is an explicit user choice and validity remains reported.

## 13.4 Noise and weighting decomposition

At each profile, noiseless OLS with fixed order and alpha, detector-weighted noiseless, and noisy detector-limited reconstructions are compared. The weighting change is $E_{\text{weighted},0} - E_{\text{OLS},0}$; the stochastic change is $\langle E_{\text{noisy}} \rangle - E_{\text{weighted},0}$. These are changes in nonlinear error metrics, not additive variance components. Signed changes may be negative in a realization; default noise count is one.

## 13.5 Basis-order diagnostic

The first source layout at $\ell_n/R = 0.15$ scans requested orders 2–8 using one shared detector-limited ensemble and the globally selected alpha. It reports density/scale-length errors, used order, rank,

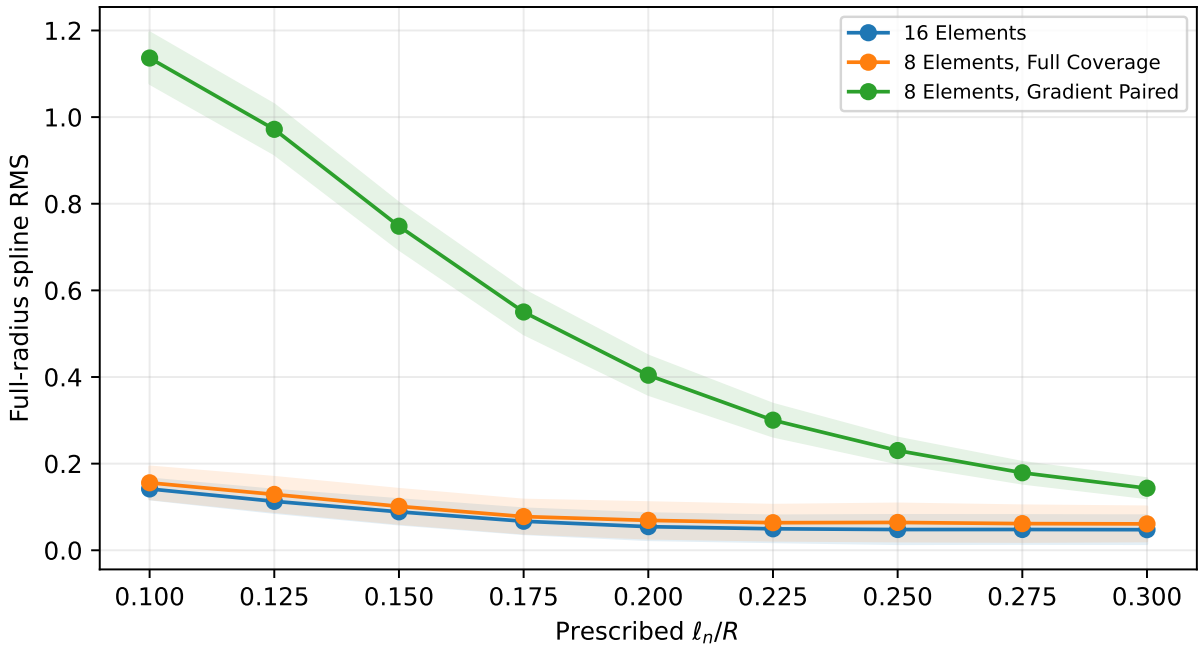



Figure 8: Full-radius density RMS for three source layouts; shaded bands show one sample standard deviation over 1000 realizations. Each layout uses its globally trained fixed estimator.

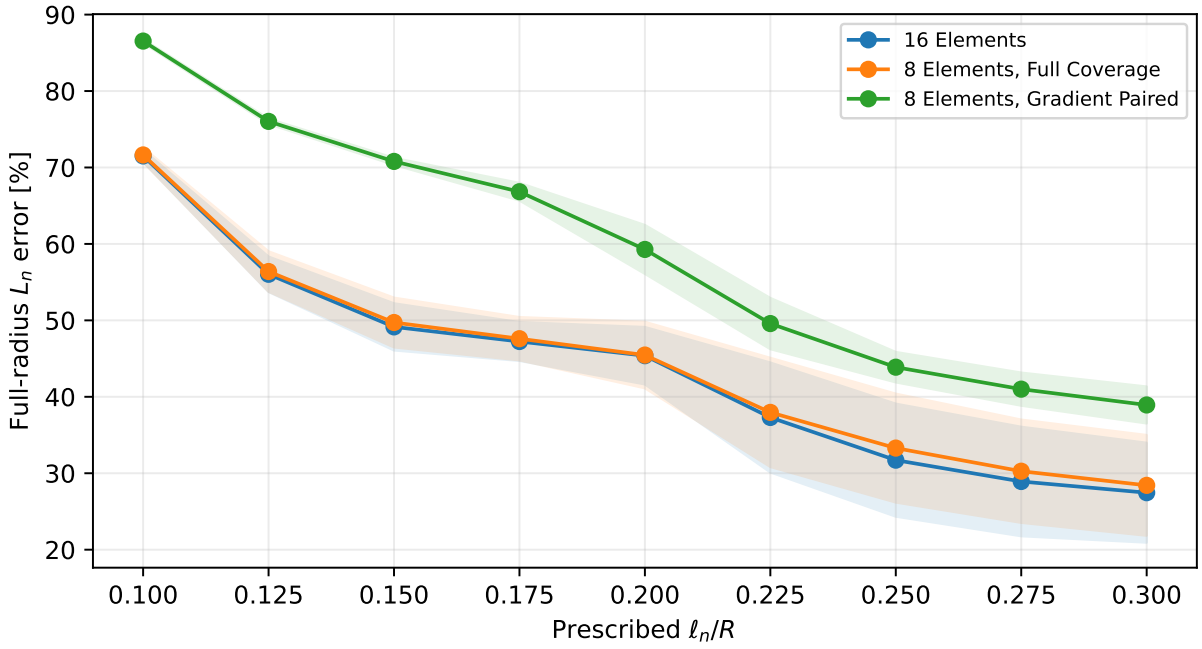



Figure 9: Corresponding bounded local-scale-length RMS. The percentage scale represents $100E_L$, not an unbounded relative error in $L_n$.

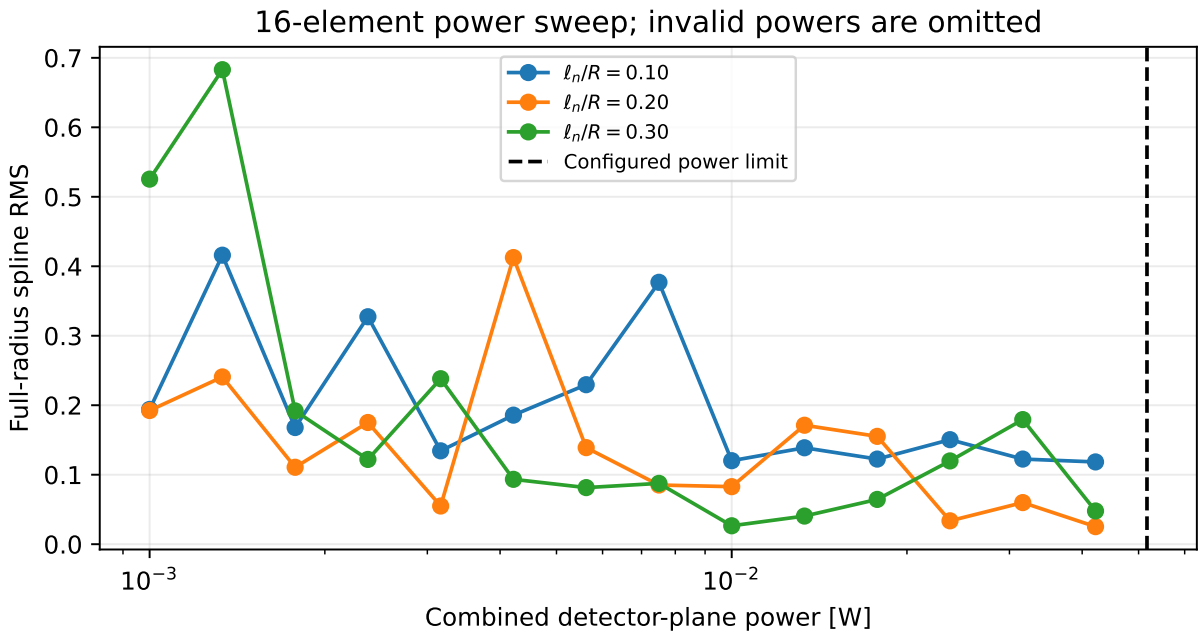



Figure 10: Baseline 16-element power sweep at the source's one-realization default. Curves stop at the configured typical-linearity operating ceiling. More power cannot cure basis/sampling bias.

and conditioning. Its best order at a single profile is not substituted for the globally trained estimator. The diagnostic also records the lowest order within the configured relative tolerance (default 1 percent) of its best score in `basis_selection.json`.

# 14 Configuration and public API reference

## 14.1 Editable SI configuration

`DesignConfig` has nested plasma, beam, port, sensor, heterodyne, detector, inversion, spline, noise, numerics, and plotting dataclasses. Every factory returns fresh mutable objects, so one study condition does not mutate another run's baseline. Validation rejects invalid sizes, nonfinite parameters, illegal power fractions, unsupported models, and inconsistent selections.

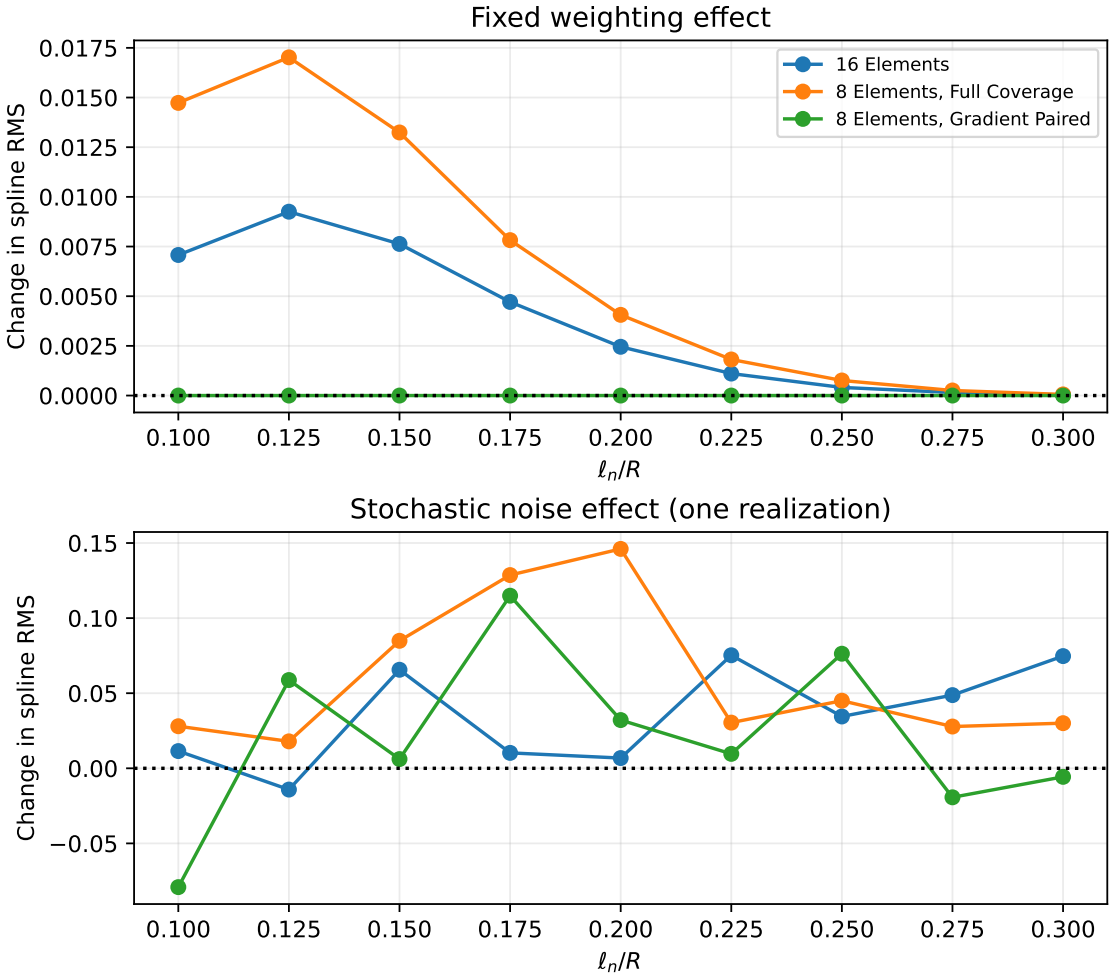



Figure 11: Fixed-estimator error changes for three representative layouts. The one-realization stochastic panel is diagnostic rather than a robust variance estimate.

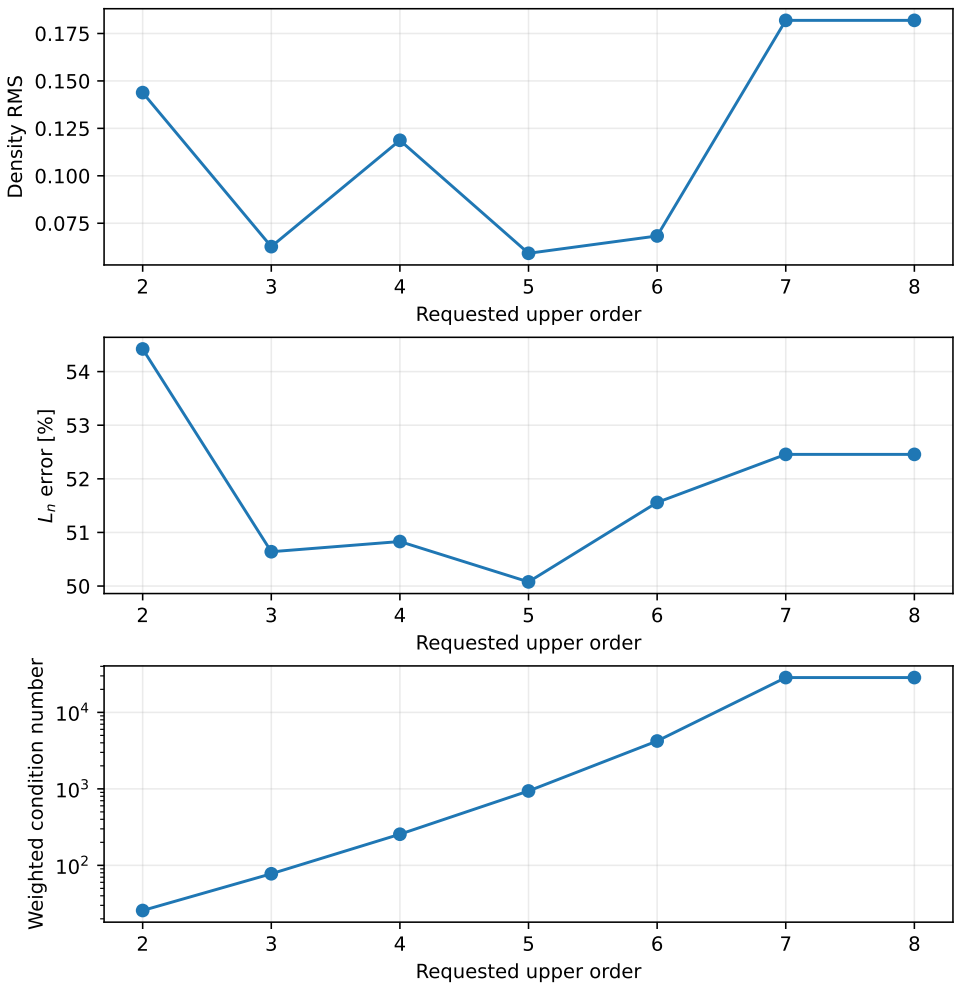



Figure 12: Baseline order diagnostic with fixed alpha. High orders expose conditioning and rank limitations; requested order 8 is reduced to an independent prefix for the centered array.

| Representative field | Default / meaning |
|---|---|
| `plasma.radius_m` | 0.10 m |
| `plasma.scale_length_m` | 0.025 m |
| `beam.waist_m` | 0.037 m |
| `magnification` | 0.057, signed |
| `heterodyne.wavelength_m` | $1.55 \times 10^{-6}$ m |
| `heterodyne.total_power_W` | 0.05 W |
| `heterodyne.bandwidth_Hz` | $10^6$ Hz |
| `inversion.upper_order` | 3 |
| `inversion.weighting_alpha` | 1 |
| `spline.lambda_value` | None: per-sample GCV |
| `noise.realizations/seed` | 1 / 1 |
| `noise.batch_size` | 256 |
| `numerics.element_order` | 20 per direction |
| `numerics.line_points/radial_points` | 1000 / 1000 |
| `plots.show/figure_format` | False / pdf |

The old README's 1 mW statement differs from the actual source's 50 mW; this release follows the source. Former main-script overrides (0.02 m profile, channels 5–12, 1000 samples) are optional examples, not hidden defaults.

### 14.2 Library interfaces and array layouts

`Workspace(cfg)` prepares geometry and bounded keyed caches. `make_measurements(cfg, workspace,normals)` returns measured ensembles without inversion; `reconstruct(ensemble, upper_order,alpha)` refits them; `simulate(cfg,workspace)` composes the two. This separation guarantees that candidate comparisons need not redraw noise. `retain_profiles=False` keeps summaries/errors without full profile ensembles. `compute_scale=False` skips derivative metrics when a tuning score does not need them.

Observations have shape $(N_{\mathrm{mc}}, N_{\mathrm{ch}})$, coefficients $(N_{\mathrm{mc}}, N_b)$, and profiles $(N_{\mathrm{mc}}, N_r)$: realizations are the first axis. Optional external normals have shape $(N_{\mathrm{mc}}, 2, N_{\mathrm{ch}})$ and require detector-limited mode. `noise.batch_size` bounds dense reconstruction/spline temporaries. A full single-run ensemble intentionally costs $O(N_{\mathrm{mc}} N_r)$ storage; metrics-only studies omit those full dense arrays.

The result dictionary contains basis/matrices, requested/used orders, rank and conditioning, coefficients, per-sample penalties, truth, sampled grid/positions, first/mean/all profiles, per-sample errors, and summaries. The attached ensemble retains geometry, measurements, detector data, specs, limits, and I/Q arrays. The full measurement CSV preserves selection/masks, image, object, and effective coordinates, transmission/power fractions, peak power and linearity, carrier/noise ASDs and SNRs, phase/coherence, linearized differences, and first observations. Detector diagnostics are computed even for noiseless modes for useful comparison, while their solve/noise behavior follows the requested mode.

### 14.3 Modules and extension boundary

`configuration.py` Configuration schema, validation, and loading.

`model.py` Geometry, optical forward model, and detector physics.

`reconstruction.py` Measurement ensembles and density inversion.

`splines.py` Symmetric penalized spline operators.

`metrics.py` Density and local scale-length errors.

`studies.py` Global training and fixed-estimator sweeps.

`outputs.py` Saved records and content-keyed tuning caches.

`plotting.py` Shared scientific figures and exports.

`entrypoints.py` Individual-script orchestration.

`native.py` validates contiguous float64 arrays and calls the compiled `_core` module. The four C routines integrate density, integrate basis, form phasors, and recover noisy I/Q phase. Python owns all buffers; C allocates no persistent memory and returns checked integer status. Complex buffers use separate real/imaginary arrays for portable MSVC support.

## 15 Efficiency and benchmark interpretation

Geometry, complete-beam normalization, and finite-area weights are constructed once per layout. Maximum-order basis integrals are cached; lower orders select columns. Profile phasors are keyed by plasma/wavelength/quadrature settings and reused across powers and noise modes. The inverse operator and spline matrices are prepared once for a given candidate/geometry, then applied to batches. The GCV penalty choice remains per realization. C recurrence and streaming avoid many trigonometric evaluations and large integration temporaries.

Scripts set the BLAS thread limit to one because these inverse systems are small; unrestricted thread startup can dominate them. NumPy/SciPy provide optimized linear algebra rather than introducing a separate user-built BLAS library. Library callers can choose their own threading policy.

Separate scripts reuse global tuning through JSON cache keys containing physics, optimization settings, package/dependency versions, actual Python-module hashes, and the compiled-extension hash. No pickle is loaded. A changed model/build invalidates the key. Caches have bounded in-memory sizes; changing geometry requires a new Workspace and is checked explicitly.

For 160 unique impacts and 1000 integration nodes on the validation machine, median density-kernel times were about 1.15 ms in C versus 2.00 ms in the same-rule NumPy reference (1.73 times faster). The order-0–8 basis bank used 4.94 ms versus 21.55 ms (4.36 times faster). Relative differences were about $3 \times 10^{-15}$. These are local kernel timings, not a MATLAB comparison or a guaranteed whole-program speedup. MATLAB runtime was unavailable.

## 16 Validation and numerical convergence

The executed tests cover analytic constant and Gaussian Abel transforms, C versus direct cosine and trapezoidal integration, positive/negative magnification, both reference modes, full tensor versus reduced phasors, zero/weak/wrapped phases, power/noise identities, safety on unselected elements,

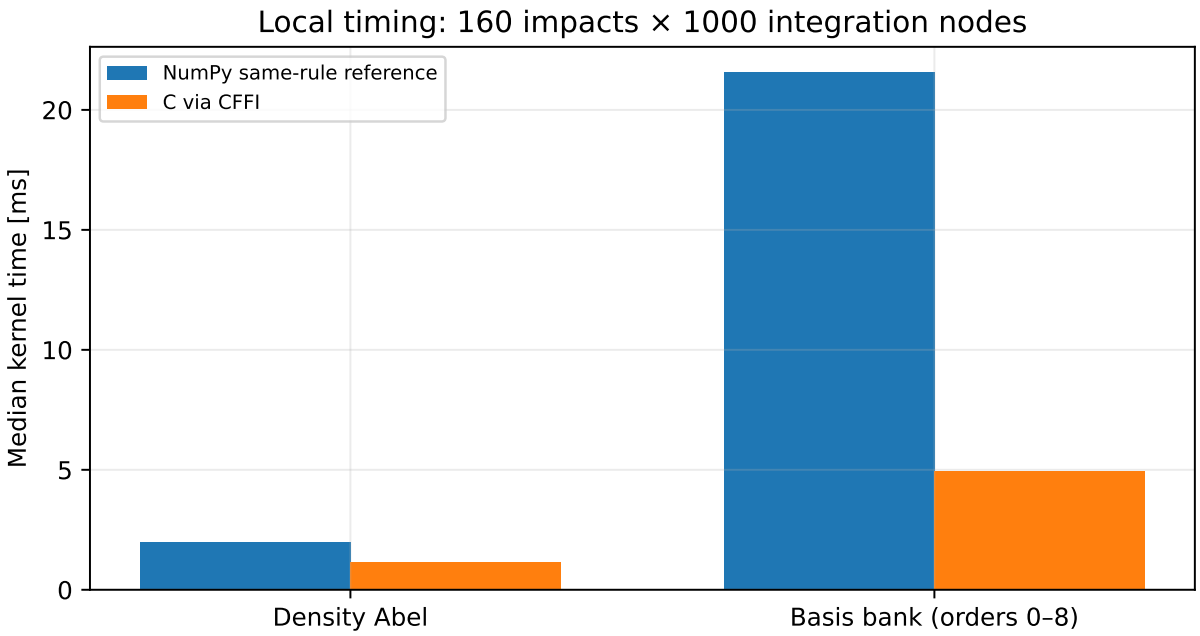



Figure 13: Median of nine local kernel timing samples after warm-up. Both implementations use the same impact parameters, quadrature rule, and precision.

geometry-cache rejection, all sensor/layout models, spline/GCV equivalence, independent weighted solves, batch invariance, I/Q statistics, noise modes, rank protection, metric conventions, invalid native inputs, and all study workflows. At preparation, 31 tests passed.

All five full scientific scripts executed with the source's default grids and sample counts on Linux x86_64, Python 3.12. The scale-length study took roughly 8.9 s including uncached tuning and plot/export; power/noise/basis runs reused matching training caches and took roughly 3.7/2.8/0.6 s. Concurrent load and export affect these numbers; they are execution records, not reproducible hardware specifications. Native macOS, Windows, and ARM64 execution remains untested, despite documented compiler recipes and compatible dependency wheels.

Increasing Gaussian Abel integration nodes demonstrates the expected smooth trapezoidal convergence in Figure 14. This is one convergence check, not a universal accuracy certificate. Port masks are discontinuous; increase element quadrature independently to check aperture-edge convergence. Also increase basis quadrature, radial resolution, Monte Carlo count, and the tuning search range before drawing a diagnostic-design conclusion. Derivative metrics are more sensitive than density metrics.

The independent comparisons reproduce the mathematical rules; they do not constitute a direct run of the original MATLAB application. A future side-by-side comparison should use the open spline path and identical supplied normal arrays, or compare distributions with justified statistical tolerances.

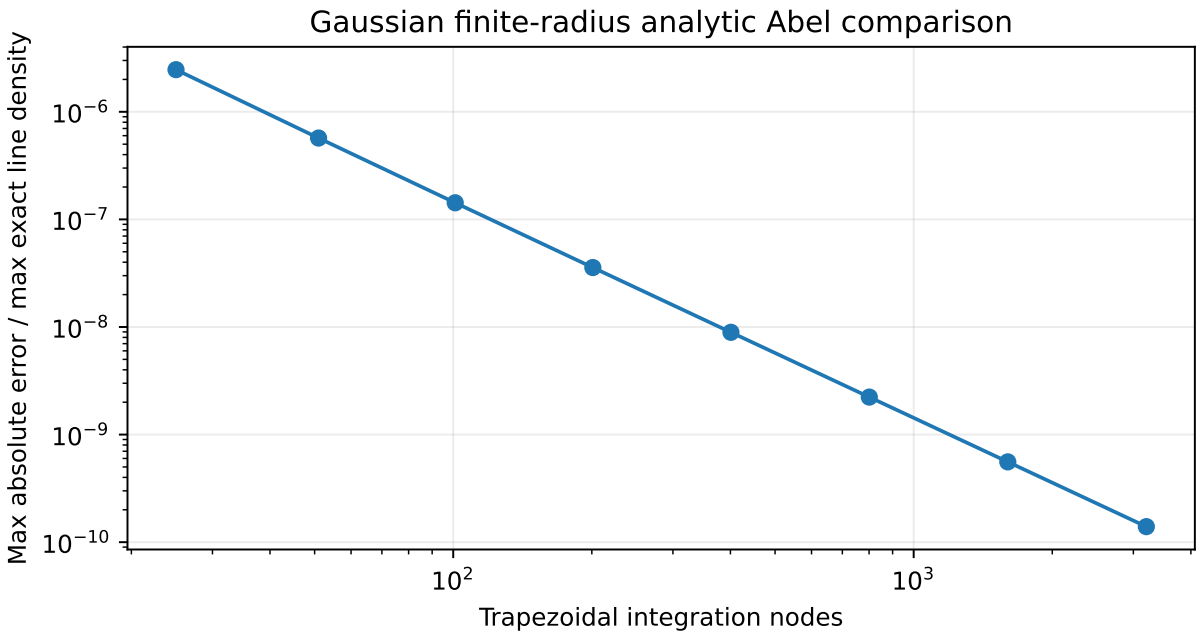



Figure 14: Maximum absolute Gaussian Abel error normalized by the maximum analytic line density over 31 impacts, including the boundary. Default profile parameters are used.

## 17 Physical limitations and troubleshooting

The optical model has ideal signed imaging at one plane, not diffraction, aberrations, depth of focus, wavefront curvature, or plasma refraction. It assumes cylindrical symmetry and $y$-uniform density. Beam mismatch and finite sampling are represented, but no general asymmetric tomography is attempted.

Vacuum calibration and I/Q axes are ideal. No temporal/spatial phase unwrapping, absolute fringe tracking, laser/AOM/LO phase noise, vibration, ADC quantization, receiver gain imbalance, or electronics saturation is included. Principal phase can wrap and finite-area cancellation can reduce the carrier. Large phase nonlinearity requires a different inverse model, not more basis coefficients.

Detector NEP and optical shot noise are included; datasheet dark current, capacitance, and resistance are not converted into additional independent noise without a receiver model. Typical linearity and fixed-wavelength detector data need experimental validation for the actual footprint/bias/electronics.

If power is rejected, inspect the critical physical element and peak-cycle power, not only selected channels. If four unique detector radii do not remain, change geometry/selection/filtering. If conditioning grows, reduce basis order or reconsider radial coverage. Outer-radius error can be extrapolation-driven; compare sampled and full-radius metrics. Low-SNR I/Q distributions can become wrapped/non-Gaussian, making analytic phase uncertainties unreliable.

For build/import errors, verify the selected Poetry Python, compiler PATH and target architecture, then rebuild with the README command. Compiled binaries cannot be copied between operating systems/interpreters. A failed native import is explicit; no silent slow fallback changes the computation.

# 18 Versioning, license, and scientific citation

The authoritative version is `project.version` in `pyproject.toml`, queried through installed metadata. Keep `version.txt`, the change history, and this manual consistent when releasing. Version 0.1.0 is the first scientific Python/C release in a separate series from MATLAB 5.0. The code is MIT licensed; dependencies retain their own licenses. Cite the code version/configuration, original inversion and measurement literature, and exact detector data used.

## References


[1] E. M. Davis, *A combined phase contrast imaging and heterodyne interferometer system for multiscale fluctuation measurements on DIII-D*, Ph.D. thesis, MIT, 2018. `https://hdl.handle.net/1721.1/115666`.

[2] Hamamatsu Photonics K.K., *InGaAs PIN photodiode arrays, G12430 series*, manufacturer datasheet KIRD1124E. `https://www.hamamatsu.com/content/dam/hamamatsu-photonics/sites/documents/99_SALES_LIBRARY/ssd/g12430_series_kird1124e.pdf`.

[3] G. Pretzler, "A New Method for Numerical Abel-Inversion," *Zeitschrift für Naturforschung A* 46, 639–641 (1991). DOI: 10.1515/zna-1991-0715.

[4] CFFI project, *Preparing Wrapper Modules* and *Using the ffi/lib Objects*. `https://cffi.readthedocs.io/en/stable/`.

[5] SciPy project, *BSpline* and *scipy.linalg.lstsq* reference documentation. `https://docs.scipy.org/doc/scipy/reference/`.

[6] Original supplied heterodyne density-reconstruction MATLAB source snapshot and documentation, version label 5.0. `https://github.com/sebasgomez17/heterodyne_density_reconstruction`.